

Student Number/Name: _____



Caringbah High School

2020

Preliminary Course
Assessment Task 3

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- NESA-approved calculators may be used
- A reference sheet is provided
- In Questions 11 - 33, show relevant mathematical reasoning and/or calculations

Total marks – 75

Section I - 10 marks (pages 2 - 5)

- Attempt Questions 1 - 10
- Mark your answers on the answer sheet provided. Detach the sheet and write your name on it.
- Allow about 15 minutes for this section

Section II - 65 marks (pages 6 - 15)

- Attempt Questions 11 - 33
- Write your answers in the space provided

Marker's Use Only									
Section I	Section II							Total	
Q 1-10	Q11 - 15	Q16 - 19	Q20 - 23	Q24 - 26	Q27 - 29	Q30 - 31	Q32 - 33		
/10	/10	/11	/10	/10	/9	/7	/8	/75	%

Section I - 10 marks**Attempt Questions 1-10****Allow about 15 minutes for this section**Use the multiple-choice answer sheet for Questions 1–10

1. What is 0.0042359 written in scientific notation correct to four significant figures?

(A) 4.2×10^{-3}
(B) 4.236×10^{-3}
(C) 4.2×10^{-4}
(D) 4.236×10^{-4}

2. If a and b are two real numbers and $a, b \neq 0$, what is the gradient of the line joining $A(a, b)$ and $B(3a, -3b)$?

(A) $-\frac{2b}{a}$
(B) $-\frac{a}{2b}$
(C) $\frac{b}{a}$
(D) $\frac{a}{b}$

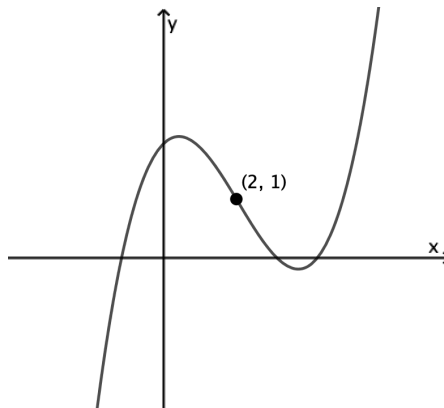
3. What are the solutions to $\sqrt{3} \tan x = -1$ for $0 \leq x \leq 2\pi$?

(A) $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$
(B) $\frac{5\pi}{6}$ and $\frac{11\pi}{6}$
(C) $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$
(D) $\frac{2\pi}{3}$ and $\frac{5\pi}{3}$

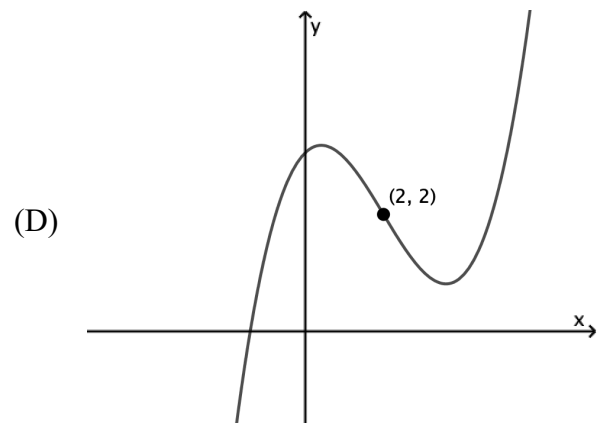
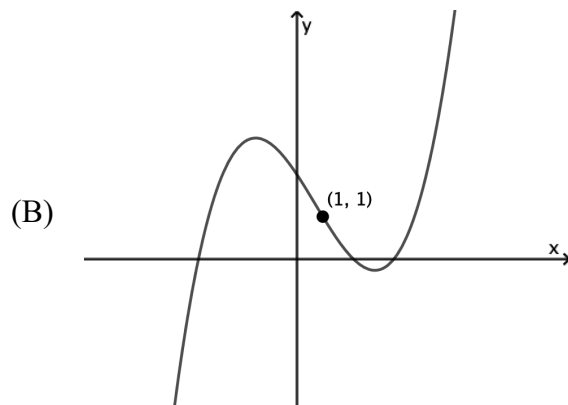
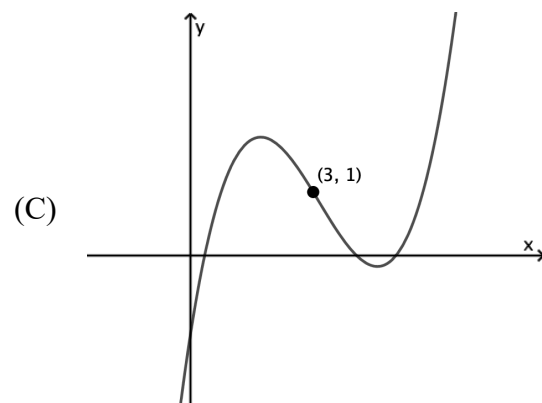
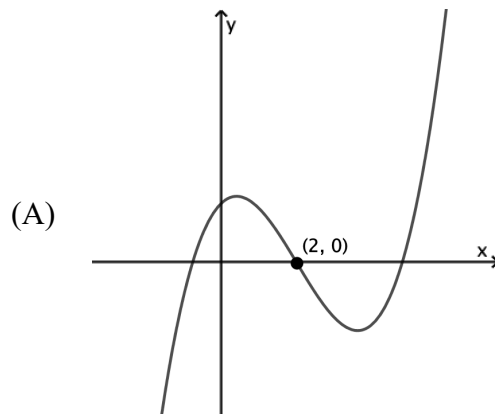
4. What is the domain of the function $f(x) = \sqrt{16 - x^2}$?

- (A) $x < 4$
- (B) $x \leq 4$
- (C) $-4 < x < 4$
- (D) $-4 \leq x \leq 4$

5. The diagram below shows the graph of $y = f(x)$:

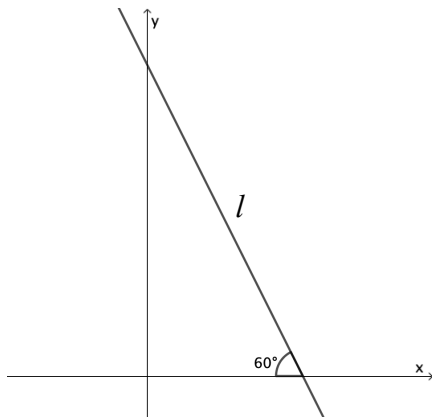


Which of these graphs represents $y = f(x - 1)$?



7. If $a = e^x$ what is the value of $\log_e \frac{1}{a^2}$?
- (A) $-x^2$
- (B) $-2x$
- (C) $2x$
- (D) x^2
6. What are the solutions of the equation $|2x - 3| = 6$?
- (A) $x = \frac{3}{2}$ or $x = -\frac{3}{2}$
- (B) $x = \frac{3}{2}$ or $x = \frac{9}{2}$
- (C) $x = \frac{9}{2}$ or $x = -\frac{3}{2}$
- (D) $x = \frac{3}{2}$ or $x = -\frac{9}{2}$
8. If $5\sqrt{2} - \sqrt{8} + \sqrt{32} = \sqrt{x}$, what is the value of x ?
- (A) 130
- (B) 98
- (C) 26
- (D) 7
9. Which of the following is the derivative of $(x^2 - 7)^4$?
- (A) $4(x^2 - 7)^3$
- (B) $4(2x - 7)^3$
- (C) $8x(x^2 - 7)^3$
- (D) $8x(2x - 7)^3$

10. The diagram shows the line l .



What is the slope of the line l ?

- (A) $\frac{1}{\sqrt{3}}$
- (B) $-\frac{1}{\sqrt{3}}$
- (C) $\sqrt{3}$
- (D) $-\sqrt{3}$

Section II

65 marks

Attempt Questions 11–33

Allow about 1 hours 45 minutes for this section

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your responses should include relevant mathematical reasoning and/or calculations.
-

Question 11 (1 mark)

Express 495° in radians. Leave your answer in terms of π

1

.....

Question 12 (2 marks)

Rationalise the denominator of $\frac{-3}{\sqrt{4+5}}$

2

.....

.....

.....

.....

.....

.....

.....

Question 13 (2 marks)

Consider $f(x) = \frac{10}{x}$ and $g(x) = 2x + 8$ what are the values of x for which $f(x) = g(x)$?

2

.....

.....

.....

.....

.....

.....

Question 14 (2 marks)

Simplify fully $\frac{6x+18}{9(x^2+4x+4)} \times \frac{x^2-4}{x+3}$

2

.....

.....

.....

.....

.....

.....

Question 15 (3 marks)

Solve for x : $4^x - 9 \times 2^x + 8 = 0$

3

.....

.....

.....

.....

.....

.....

Question 16 (2 marks)

Find the equation of the tangent to the curve $y = 5x^3 - 4x + 3$ at the point $(-1, 2)$.

2

.....

.....

.....

.....

.....

Question 17 (2 marks)

Two identical *biased* coins are tossed at the same time. The probability that both coins land on heads is 0.16. Find the probability that:

- (a) Both coins land on tails.

1

.....

.....

.....

- (b) At least one coin lands on heads.

1

.....

.....

.....

Question 18 (5 marks)

Differentiate the following with respect to x . Simplify your answers.

- (a) $y = (x\sqrt{x} + 1)^3$

2

.....

.....

.....

- (b) $y = \frac{e^{2x}}{x^2 - 3}$

3

.....

.....

.....

.....

.....

.....

Question 19 (2 marks)

Show that $f(x) = 2x^3 + 4x$ is an odd function.

2

.....

.....

.....

.....

Question 20 (2 marks)

Using the first principles formula, $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ when $f(x) = x^2 - 2x$

2

.....

.....

.....

.....

.....

.....

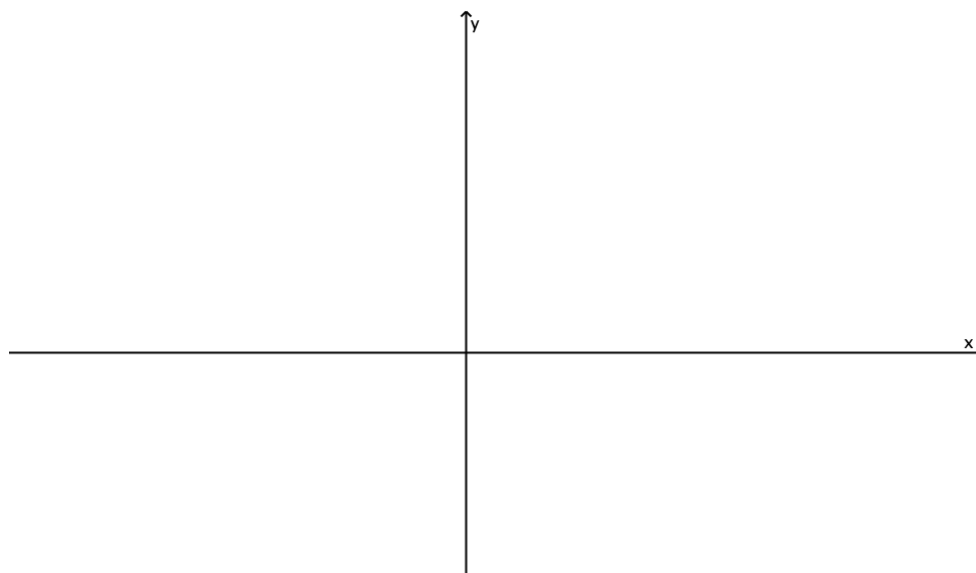
.....

.....

Question 21 (2 marks)

Sketch the graph of $f(x) = 1 - \log_e x$ showing the x -intercept and the equation of the asymptote.

2



Question 22 (4 marks)

The point $P(4, 4)$ lies on the curve $y = -x^2 + 3x + 8$. The normal to the curve at P intersects the curve again at point Q .

(a) Show that the equation of the normal at P is $-x + 5y = 16$ 2

.....

.....

.....

.....

.....

(b) Find the coordinates of Q . 2

.....

.....

.....

.....

.....

.....

Question 23 (2 marks)

Find the coordinates of the stationary point(s) on the curve $y = x(\sqrt{x} - 3)$ 2

.....

.....

.....

.....

.....

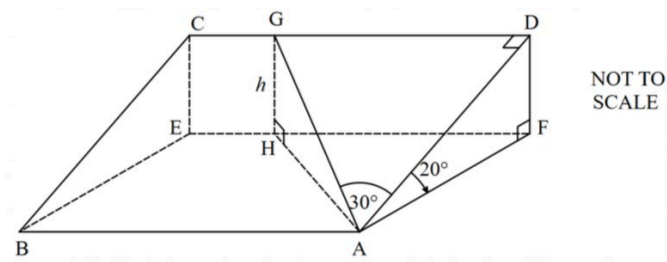
.....

.....

.....

Question 24 (4 marks)

‘Family hill’, a grassed spectating area at Shark Park makes an angle of 20° with the horizontal. In the representation below, the path AG makes an angle of 30° with the line of greatest slope, and $DF = GH = h$. This is shown in the diagram below.



- (a) Show that $AD = \frac{h}{\sin 20}$ 1

.....

.....

.....

.....

- (b) Show that $AG = \frac{h}{\sin 20 \cos 30}$ 1

.....

.....

.....

.....

.....

- (c) Find the angle of inclination of the path AG to the horizontal. Give your answer correct to the nearest degree. 2

.....

.....

.....

.....

.....

Question 25 (4 marks)

(a) Show that $3 \cos x + 3 = -2 \sin^2 x$ can be written as $2 \cos^2 x - 3 \cos x - 5 = 0$ **2**

.....

.....

.....

.....

.....

(b) Hence or otherwise, solve $3 \cos x + 3 = -2 \sin^2 x$ for $0 \leq x \leq 2\pi$. **2**

.....

.....

.....

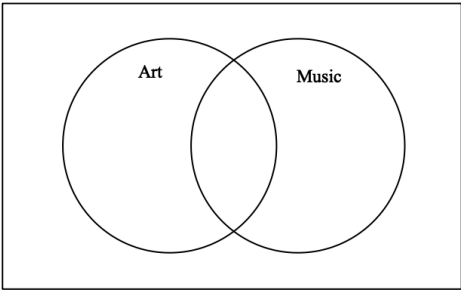
.....

.....

Question 26 (3 marks)

A class has 25 students. 14 of them study Art and 12 study Music. 5 students study neither Art nor Music. Two students are chosen at random from the class.

(a) Draw a Venn diagram to represent the above information showing the number of students studying Art but not Music, and the number of students studying both Art and Music. **1**



(b) Find the probability that both students chosen study Art. **2**

.....

.....

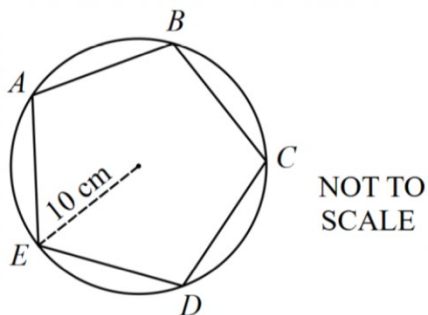
.....

.....

Question 27 (4 marks)

4

A regular pentagon $ABCDE$ is inscribed in a circle of radius 10 cm, as shown below.



Find the perimeter of the pentagon, correct to 3 significant figures.

.....

.....

.....

.....

.....

.....

Question 28 (1 mark)

1

Find the values of x where the graph of the function $f(x) = \frac{1}{x^2-4}$ is discontinuous.

.....

.....

.....

Question 29 (3 marks)

3

Solve the equation $\log_2(x + 1) + \log_2(x - 1) = 3$

.....

.....

.....

.....

Question 30 (4 marks)

In a block of 40 apartments, when the weekly rent for each apartment is \$570, all 40 apartments are rented out. For each increase of \$20 in the weekly rent, the number of apartments rented out decreases by one. Each empty apartment incurs a weekly cost of \$70 for the block owner.

- (a) Show that if the weekly rent is increased by n lots of \$20 ($n = 0, 1, 2, 3 \dots$) then the total weekly profit $\$P$ from the apartments is given by $P = 22\,800 + 160n - 20n^2$ 2

.....

.....

.....

.....

.....

- (b) Find the maximum weekly profit.2

.....

.....

.....

.....

.....

Question 31 (3 marks)

Prove that $\frac{1+\sin \theta}{1-\sin \theta} - \frac{1-\sin \theta}{1+\sin \theta} = 4 \sec \theta \tan \theta$ 3

.....

.....

.....

.....

.....

.....

.....

.....

Question 32 (5 marks)

The energy, E Megajoules, radiated by an earthquake is given by the formula $E = k \times 10^{1.5M}$ where M is the magnitude of the earthquake on the Richter scale and k is a constant. An earthquake of magnitude 1 radiates 2 megajoules of energy.

- (a) Show that $\log_{10} E = \log_{10} 2 + 1.5(M - 1)$ **3**

.....

.....

.....

.....

.....

- (b) Find the magnitude of an earthquake which radiates 2×10^6 megajoules of energy **2**

.....

.....

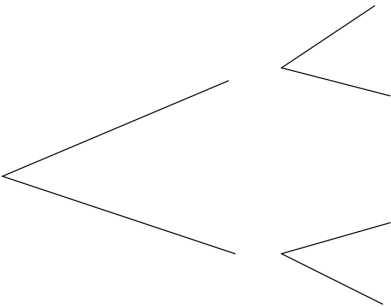
.....

.....

Question 33 (3 marks)

On his way to school each day Ozzie catches a train and a bus. The train is late 10% of the time and the bus is late 20% of the time. If either the train or the bus is late then Ozzie is late for school.

- (a) Draw a probability tree diagram to represent the above information. **1**



- (b) Find the probability that Ozzie is late for school on any given day. **2**

.....

.....

Extra Writing Space

If you use this space, clearly indicate which question you are answering.

[illegible]

Student Name: _____

Multiple choice answer page. Fill in either A, B, C or D for questions 1-10.

This page must be handed in with your answer booklets

1.	
2.	
3.	
4.	
5.	

6.	
7.	
8.	
9.	
10.	

Student Number/Name: Solutions.



Caringbah High School

2020

Preliminary Course
Assessment Task 3

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- NESA-approved calculators may be used
- A reference sheet is provided
- In Questions 11 - 33, show relevant mathematical reasoning and/or calculations

Total marks – 75

Section I - 10 marks (pages 2 - 5)

- Attempt Questions 1 - 10
- Mark your answers on the answer sheet provided. Detach the sheet and write your name on it.
- Allow about 15 minutes for this section

Section II - 65 marks (pages 6 - 15)

- Attempt Questions 11 - 33
- Write your answers in the space provided

Marker's Use Only									
Section I	Section II							Total	
Q 1-10	Q11 - 15	Q16 - 19	Q20 - 23	Q24 - 26	Q27 - 29	Q30 - 31	Q32 - 33		
/10	/10	/11	/10	/10	/9	/7	/8	/75	%

Section I - 10 marks**Attempt Questions 1-10****Allow about 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1–10

1. What is 0.0042359 written in scientific notation correct to four significant figures?

(A) 4.2×10^{-3}

(B) 4.236×10^{-3}

(C) 4.2×10^{-4}

(D) 4.236×10^{-4}

B

2. If a and b are two real numbers and $a, b \neq 0$, what is the gradient of the line joining $A(a, b)$ and $B(3a, -3b)$?

(A) $-\frac{2b}{a}$

(B) $-\frac{a}{2b}$

(C) $\frac{b}{a}$

(D) $\frac{a}{b}$

$$m = \frac{-3b - b}{3a - a}$$

$$= \frac{-4b}{2a}$$

$$= \frac{-2b}{a}$$

A

3. What are the solutions to $\sqrt{3} \tan x = -1$ for $0 \leq x \leq 2\pi$?

(A) $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$

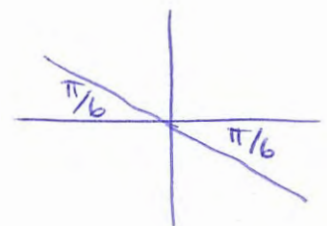
(B) $\frac{5\pi}{6}$ and $\frac{11\pi}{6}$

(C) $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$

(D) $\frac{2\pi}{3}$ and $\frac{5\pi}{3}$

$$\tan x = -\frac{1}{\sqrt{3}}$$

$$x = \frac{5\pi}{6}, \frac{11\pi}{6}$$



B

4. What is the domain of the function $f(x) = \sqrt{16 - x^2}$?

(A) $x < 4$

(B) $x \leq 4$

(C) $-4 < x < 4$

(D) $-4 \leq x \leq 4$

$$16 - x^2 = 0$$

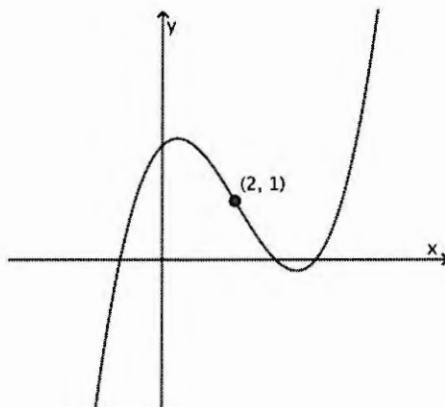
$$x^2 = 16$$

$$x = \pm 4$$

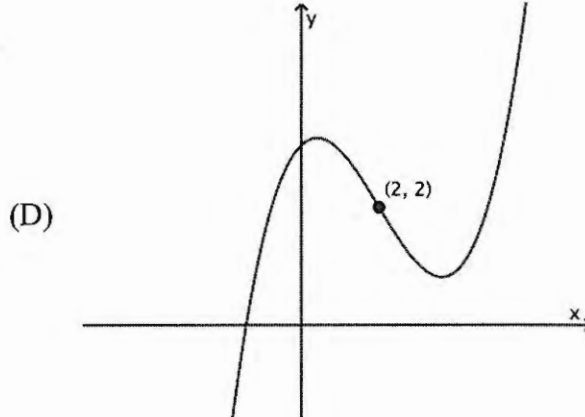
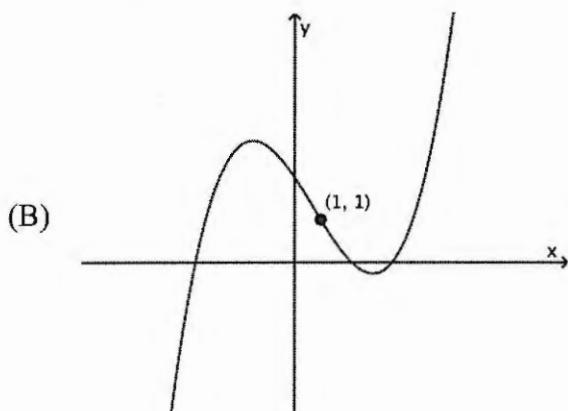
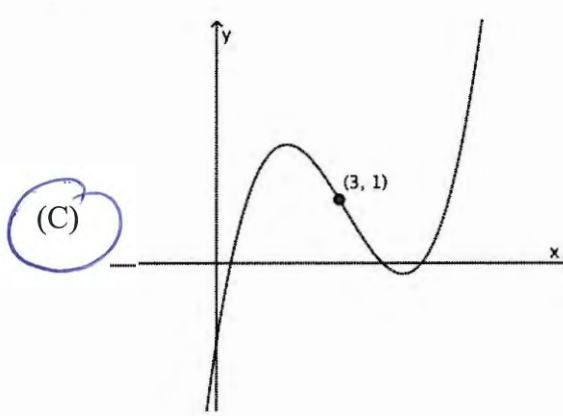
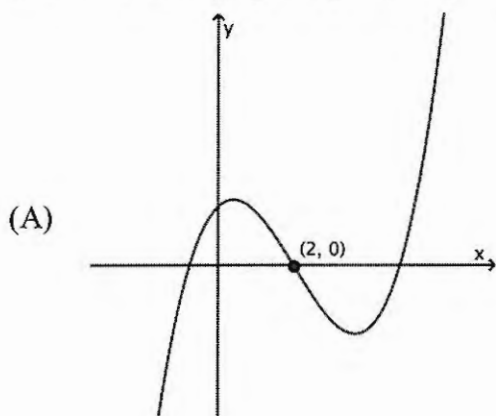
$$-4 \leq x \leq 4$$

D

5. The diagram below shows the graph of $y = f(x)$:



Which of these graphs represents $y = f(x - 1)$?



C

7. If $a = e^x$ what is the value of $\log_e \frac{1}{a^2}$?

- (A) $-x^2$
- ☒ (B) $-2x$
- (C) $2x$
- (D) x^2

$$\begin{aligned}\log_e \frac{1}{e^{2x}} \\&= \log_e 1 - \log_e e^{2x} \\&= 0 - 2x\end{aligned}$$

B

6. What are the solutions of the equation $|2x - 3| = 6$?

- (A) $x = \frac{3}{2}$ or $x = -\frac{3}{2}$
- (B) $x = \frac{3}{2}$ or $x = \frac{9}{2}$
- ☒ (C) $x = \frac{9}{2}$ or $x = -\frac{3}{2}$
- (D) $x = \frac{3}{2}$ or $x = -\frac{9}{2}$

$$\begin{aligned}2x - 3 &= 6 \\x &= \frac{9}{2}\end{aligned}$$

$$\begin{aligned}2x - 3 &= -6 \\x &= -\frac{3}{2}\end{aligned}$$

C

8. If $5\sqrt{2} - \sqrt{8} + \sqrt{32} = \sqrt{x}$, what is the value of x ?

- (A) 130
- ☒ (B) 98
- (C) 26
- (D) 7

$$\begin{aligned}5\sqrt{2} - 2\sqrt{2} + 4\sqrt{2} \\&= 7\sqrt{2} \\&= \sqrt{49 \times 2} \\&= \sqrt{98}\end{aligned}$$

B

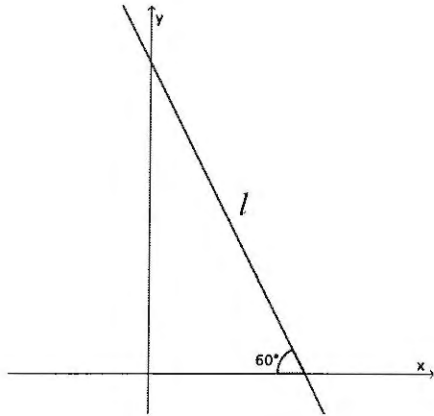
9. Which of the following is the derivative of $(x^2 - 7)^4$?

- (A) $4(x^2 - 7)^3$
- (B) $4(2x - 7)^3$
- ☒ (C) $8x(x^2 - 7)^3$
- (D) $8x(2x - 7)^3$

$$\begin{aligned}y' &= 4(x^2 - 7)^3 \times 2x \\&= 8x(x^2 - 7)^3\end{aligned}$$

C

10. The diagram shows the line l .



What is the slope of the line l ?

- (A) $\frac{1}{\sqrt{3}}$
- (B) $-\frac{1}{\sqrt{3}}$
- (C) $\sqrt{3}$
- (D) $-\sqrt{3}$

$$\tan \theta = m$$

$$\tan (120^\circ) = -\sqrt{3}$$

Section II

65 marks

Attempt Questions 11–33

Allow about 1 hours 45 minutes for this section

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (1 mark)

Express 495° in radians. Leave your answer in terms of π

1

$$2.75\pi \quad \text{or} \quad \frac{11\pi}{4} \quad \checkmark$$

Question 12 (2 marks)

Rationalise the denominator of $\frac{-3}{\sqrt{4}+5}$

2

$$= \frac{-3}{\sqrt{4}+5} \times \frac{\sqrt{4}-5}{\sqrt{4}-5}$$

$$= \frac{-3\sqrt{4}+15}{4-25}$$

$$= \frac{-3\sqrt{4}+15}{-21} \quad \checkmark$$

$$= \frac{\sqrt{4}-5}{7}$$

$$\text{OR} \quad \frac{-5+\sqrt{4}}{7} \quad \checkmark \quad \text{OR} \quad -\frac{5-\sqrt{4}}{7}$$

Question 13 (2 marks)

Consider $f(x) = \frac{10}{x}$ and $g(x) = 2x + 8$ what are the values of x for which $f(x) = g(x)$?

2

$$\frac{10}{x} = 2x + 8$$

$$x = -5 \quad \text{or}$$

$$10 = 2x^2 + 8x$$

$$x = 1 \quad \checkmark$$

$$2x^2 + 8x - 10 = 0$$

$$x^2 + 4x - 5 = 0 \quad \checkmark$$

$$(x+5)(x-1) = 0$$

Question 14 (2 marks)

Simplify fully $\frac{6x+18}{9(x^2+4x+4)} \times \frac{x^2-4}{x+3}$

2

$$\frac{6\cancel{(x+3)}}{9(x+2)^2} \times \frac{(x-2)\cancel{(x+2)}}{\cancel{x+3}} \checkmark$$

$$= \frac{2(x-2)}{3(x+2)} \checkmark$$

Question 15 (3 marks)

Solve for x : $4^x - 9 \times 2^x + 8 = 0$

3

$2^{2x} - 9 \times 2^x + 8 = 0$	$2^x = 8$	$2^x = 1$
let $u = 2^x$	$2^x = 2^3$	$2^x = 2^0$
$u^2 - 9u + 8 = 0 \checkmark$	$x = 3$	$x = 0$
$(u-8)(u-1) = 0$	or	
$u = 8, u = 1 \checkmark$	$\checkmark$	

Question 16 (2 marks)

Find the equation of the tangent to the curve $y = 5x^3 - 4x + 3$ at the point $(-1, 2)$.

2

$y' = 15x^2 - 4$	$y - 2 = 11(x + 1)$
at $x = -1$,	$y = 11x + 13$
$m = 15 - 4$	$\checkmark$
$= 11 \checkmark$	or
	$11x - y + 13 = 0$

Question 17 (2 marks)

Two identical *biased* coins are tossed at the same time. The probability that both coins land on heads is 0.16. Find the probability that:

- (a) Both coins land on tails.

1

$$\begin{array}{l|l}
 P(HH) = 0.16 & P(T) = 0.6 \\
 = P(H) \times P(H) & P(TT) = 0.6 \times 0.6 \\
 \therefore P(H) = 0.4 & = 0.36 \checkmark
 \end{array}$$

- (b) At least one coin lands on heads.

1

$$\begin{aligned}
 P(\text{at least one head}) &= 1 - P(TT) \\
 &= 1 - 0.36 \\
 &= 0.64 \checkmark
 \end{aligned}$$

Question 18 (5 marks)

Differentiate the following with respect to x . Simplify your answers.

- (a) $y = (x\sqrt{x} + 1)^3$

2

$$\begin{aligned}
 \frac{dy}{dx} &= 3(x\sqrt{x} + 1)^2 \times \frac{3}{2}x^{1/2} \checkmark \\
 &= \frac{9\sqrt{x}}{2} (x\sqrt{x} + 1)^2 \checkmark
 \end{aligned}$$

- (b) $y = \frac{e^{2x}}{x^2 - 3}$

3

$$\begin{aligned}
 u &= e^{2x} & v &= x^2 - 3 \\
 u' &= 2e^{2x} \checkmark & v' &= 2x \\
 \frac{dy}{dx} &= \frac{2e^{2x}x^2 - 6e^{2x} - 2e^{2x}x}{(x^2 - 3)^2} = \frac{2e^{2x}(x^2 - x - 3)}{(x^2 - 3)^2} \checkmark
 \end{aligned}$$

Question 19 (2 marks)

Show that $f(x) = 2x^3 + 4x$ is an odd function.

2

$$\begin{aligned}
 f(-x) &= 2(-x)^3 + 4(-x) \\
 &= -2x^3 - 4x \checkmark \\
 -f(x) &= -2x^3 - 4x \\
 &= f(-x) \therefore f(x) \text{ is odd.} \checkmark
 \end{aligned}$$

Question 20 (2 marks)

Using the first principles formula, $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ when $f(x) = x^2 - 2x$

2

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 2(x+h) - x^2 + 2x}{h} \checkmark \\
 &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 2x - 2h - x^2 + 2x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 2h}{h} \\
 &= \lim_{h \rightarrow 0} 2x + h - 2 \\
 &= 2x - 2 \checkmark
 \end{aligned}$$

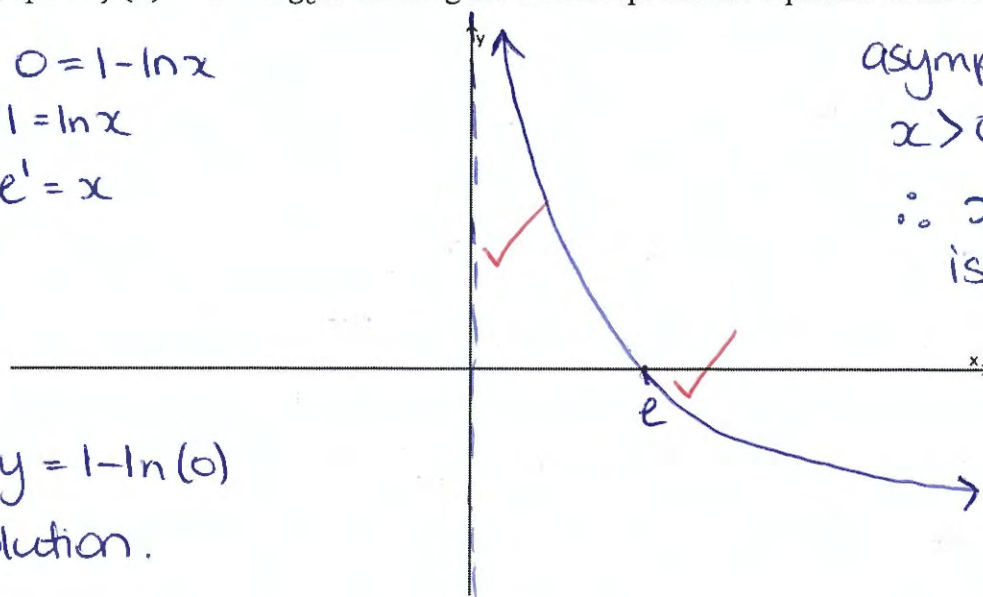
Question 21 (2 marks)

2

Sketch the graph of $f(x) = 1 - \log_e x$ showing the x -intercept and the equation of the asymptote.

$x\text{-int: } 0 = 1 - \ln x$
 $1 = \ln x$
 $e^1 = x$

$y\text{-int: } y = 1 - \ln(0)$
 no solution.



asymptote:
 $x > 0$
 $\therefore x = 0$
 is asymptote.

Question 22 (4 marks)

The point $P(4, 4)$ lies on the curve $y = -x^2 + 3x + 8$. The normal to the curve at P intersects the curve again at point Q .

- (a) Show that the equation of the normal at P is $-x + 5y = 16$

2

$$\frac{dy}{dx} = -2x + 3$$

$$m_T = -2(4) + 3$$

$$= -5$$

$$m_N = \frac{1}{5} \checkmark$$

$$y - 4 = \frac{1}{5}(x - 4)$$

$$5y - 20 = x - 4$$

$$-x + 5y = 16 \checkmark$$

- (b) Find the coordinates of Q .

2

① $-x + 5y = 16$ intersects $y = -x^2 + 3x + 8$ ②

② $\rightarrow$ ①

$$-x + 5(-x^2 + 3x + 8) = 16$$

$$-x - 5x^2 + 15x + 40 = 16$$

$$5x^2 - 14x - 24 = 0$$

$$(x - 4)(5x + 6) = 0 \checkmark$$

$$x = 4 \text{ and } x = -\frac{6}{5}$$

$$\text{when } x = -\frac{6}{5}$$

$$y = \frac{74}{25} \checkmark$$

Question 23 (2 marks)

Find the coordinates of the stationary point(s) on the curve $y = x(\sqrt{x} - 3)$

2

$$\frac{dy}{dx} = \frac{3}{2}x^{1/2} - 3 \checkmark$$

$$y = x^{3/2} - 3x$$

$$\frac{3\sqrt{x} - 3}{2} = 0$$

$$\frac{3\sqrt{x}}{2} = 3$$

$$3\sqrt{x} = 6$$

$$\sqrt{x} = 2$$

$$\therefore x = 4$$

$$\text{When } x = 4$$

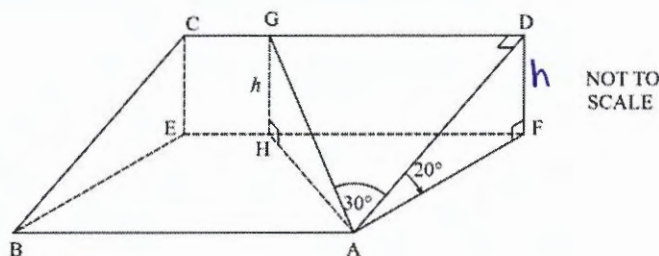
$$y = 4(2 - 3)$$

$$= -4$$

$$(4, -4) \checkmark$$

Question 24 (4 marks)

'Family hill', a grassed spectating area at Shark Park makes an angle of 20° with the horizontal. In the representation below, the path AG makes an angle of 30° with the line of greatest slope, and $DF = GH = h$. This is shown in the diagram below.



(a) Show that $AD = \frac{h}{\sin 20}$

1

$$\sin 20^\circ = \frac{h}{AD}$$

$$AD = \frac{h}{\sin 20^\circ} \checkmark$$

(b) Show that $AG = \frac{h}{\sin 20 \cos 30}$

1

$$\cos 30^\circ = \frac{AD}{AG}$$

$$AG = \frac{AD}{\cos 30^\circ}$$

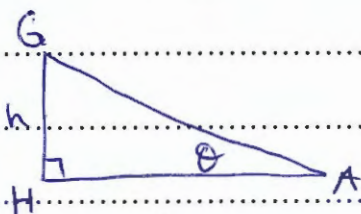
$$AD = \frac{h}{\sin 20}$$

$$AG = \frac{h}{\sin 20} \div \cos 30$$

$$AG = \frac{h}{\sin 20 \cos 30} \checkmark$$

(c) Find the angle of inclination of the path AG to the horizontal. Give your answer correct to the nearest degree.

2



$$\sin \theta = \frac{h}{AG}$$

$$\sin \theta = \cancel{h} \times \frac{\sin 20 \cos 30}{\cancel{h}} \checkmark$$

$$\sin \theta = \sin 20 \cos 30$$

$$\theta \doteq 17^\circ \checkmark$$

Question 25 (4 marks)

- (a) Show that $3 \cos x + 3 = -2 \sin^2 x$ can be written as $2 \cos^2 x - 3 \cos x - 5 = 0$

2

$$3 \cos x + 3 = -2(1 - \cos^2 x) \quad \checkmark$$

$$3 \cos x + 3 = -2 + 2 \cos^2 x$$

$$2 \cos^2 x - 3 \cos x - 5 = 0 \quad \checkmark$$

- (b) Hence or otherwise, solve $3 \cos x + 3 = -2 \sin^2 x$ for $0 \leq x \leq 2\pi$.

2

$$\text{let } u = \cos x \quad (2u-5)(u+1) = 0$$

$$2u^2 - 3u - 5 = 0 \quad u = 5/2 \quad u = -1 \quad \checkmark$$

$$\begin{array}{l} P: -10 \\ S: -3 \\ F: -5, 2 \end{array} \quad \left. \begin{array}{l} 2u^2 - 5u + 2u - 5 = 0 \\ u(2u-5) + (2u-5) = 0 \end{array} \right\} \quad \cos x = 5/2 \rightarrow \text{no sol.}$$

$$\cos x = -1$$

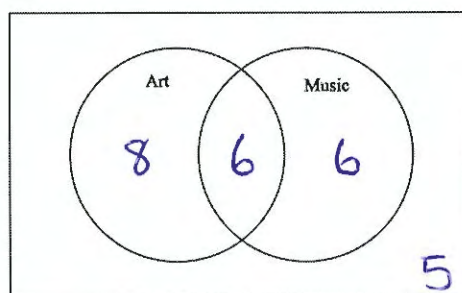
$$\underline{x = \pi} \quad \checkmark$$

Question 26 (3 marks)

A class has 25 students. 14 of them study Art and 12 study Music. 5 students study neither Art nor Music. Two students are chosen at random from the class.

- (a) Draw a Venn diagram to represent the above information showing the number of students studying Art but not Music, and the number of students studying both Art and Music.

1



- (b) Find the probability that both students chosen study Art.

2

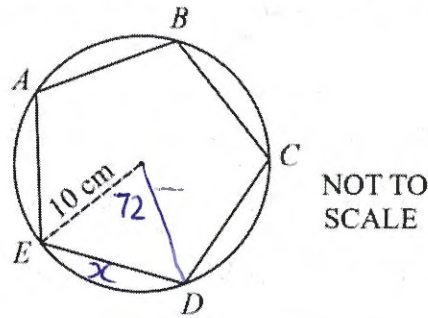
$$P(AA) = \frac{14}{25} \times \frac{13}{24} \quad \checkmark$$

$$= \frac{91}{300} \quad \text{OR } 0.303 \quad \checkmark$$

Question 27 (4 marks)

4

A regular pentagon $ABCDE$ is inscribed in a circle of radius 10 cm, as shown below.



Find the perimeter of the pentagon, correct to 3 significant figures.

$$x^2 = 10^2 + 10^2 - 2 \times 10 \times 10 \times \cos 72^\circ \checkmark$$

$$= 138.196 \dots$$

$$x \div 11.755 \checkmark$$

$$\therefore P = 5 \times 11.755 \checkmark$$

$$P \div 58.8 \text{ cm} \checkmark$$

Question 28 (1 mark)

1

Find the values of x where the graph of the function $f(x) = \frac{1}{x^2-4}$ is discontinuous.

$$x^2 - 4 = 0$$

$$x = \pm 2 \checkmark$$

Question 29 (3 marks)

3

Solve the equation $\log_2(x+1) + \log_2(x-1) = 3$

$$\log_2((x+1)(x-1)) = 3$$

$$\log_2(x^2-1) = 3 \checkmark$$

$$2^3 = x^2 - 1$$

$$x^2 = 9$$

$$x = \pm 3 \checkmark$$

however,

when $x = -3$,

$\log_2(x+1)$ AND $\log_2(x-1)$
are undefined.

$$\therefore \underline{x = 3} \checkmark$$

Question 30 (4 marks)

In a block of 40 apartments, when the weekly rent for each apartment is \$570, all 40 apartments are rented out. For each increase of \$20 in the weekly rent, the number of apartments rented out decreases by one. Each empty apartment incurs a weekly cost of \$70 for the block owner.

- (a) Show that if the weekly rent is increased by n lots of \$20 ($n = 0, 1, 2, 3 \dots$) then the total weekly profit \$ P from the apartments is given by $P = 22\,800 + 160n - 20n^2$ 2

$$\begin{aligned}
 P &= (570 + 20n) \times (40 - n) - 70n \\
 &= 22800 - 570n + 800n - 20n^2 - 70n \\
 &= 22800 + 160n - 20n^2
 \end{aligned}$$

- (b) Find the maximum weekly profit. 2

$$\begin{aligned}
 &\text{Concave down parabola } \therefore \text{Vertex is maximum} \\
 x &= \frac{-b}{2a} = \frac{-160}{-40} = 4 \\
 P &= 22800 + (160 \times 4) - (20 \times 16) = \$23120
 \end{aligned}$$

Question 31 (3 marks)

Prove that $\frac{1+\sin\theta}{1-\sin\theta} - \frac{1-\sin\theta}{1+\sin\theta} = 4 \sec\theta \tan\theta$ 3

$$\begin{aligned}
 \text{LHS} &= \frac{1+\sin\theta}{1-\sin\theta} - \frac{1-\sin\theta}{1+\sin\theta} \\
 &= \frac{(1+\sin\theta)^2 - (1-\sin\theta)^2}{(1+\sin\theta)(1-\sin\theta)} \\
 &= \frac{1 + 2\sin\theta + \sin^2\theta - 1 + 2\sin\theta - \sin^2\theta}{1 - \sin^2\theta} \\
 &= \frac{4\sin\theta}{\cos^2\theta} = 4\tan\theta \times \frac{1}{\cos\theta} \\
 &= 4\sec\theta \tan\theta = \text{RHS}
 \end{aligned}$$

Question 32 (5 marks)

The energy, E Megajoules, radiated by an earthquake is given by the formula $E = k \times 10^{1.5M}$ where M is the magnitude of the earthquake on the Richter scale and k is a constant. An earthquake of magnitude 1 radiates 2 megajoules of energy.

- (a) Show that $\log_{10} E = \log_{10} 2 + 1.5(M - 1)$

3

$$\begin{array}{l|l}
 E = k \times 10^{1.5M} & E = 2 \times 10^{1.5(M-1)} \\
 2 = k \times 10^{1.5} & \log_{10} \left(\frac{E}{2} \right) = 1.5(M-1) \checkmark \\
 \frac{2}{10^{1.5}} = k & \log_{10} E - \log_{10} 2 = 1.5(M-1) \\
 k = 2 \times 10^{-1.5} \checkmark & \log_{10} E = \log_{10} 2 + 1.5(M-1) \checkmark \\
 E = 2 \times 10^{-1.5} \times 10^{1.5M} &
 \end{array}$$

- (b) Find the magnitude of an earthquake which radiates 2×10^6 megajoules of energy

2

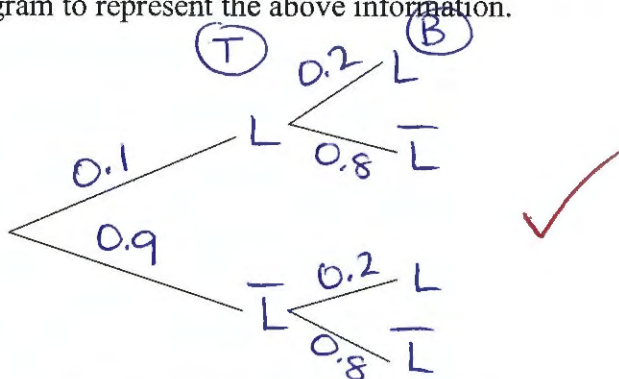
$$\begin{array}{l|l}
 \log_{10}(2 \times 10^6) = \log_{10} 2 + 1.5(M-1) & M-1 = 13.95 \\
 \log_{10} \left(\frac{2 \times 10^6}{2} \right) = 1.5(M-1) & M = 14.95 \checkmark \\
 20.93 = 1.5(M-1) \checkmark &
 \end{array}$$

Question 33 (3 marks)

On his way to school each day Ozzie catches a train and a bus. The train is late 10% of the time and the bus is late 20% of the time. If either the train or the bus is late then Ozzie is late for school.

- (a) Draw a probability tree diagram to represent the above information.

1



- (b) Find the probability that Ozzie is late for school on any given day.

2

$$\begin{aligned}
 P(L) + P(L) &= 1 - P(\bar{L}\bar{L}) \\
 &= 1 - (0.9 \times 0.8) \checkmark = 0.28 \\
 &\quad \text{or } \frac{7}{25} \checkmark
 \end{aligned}$$

End of Exam ☺